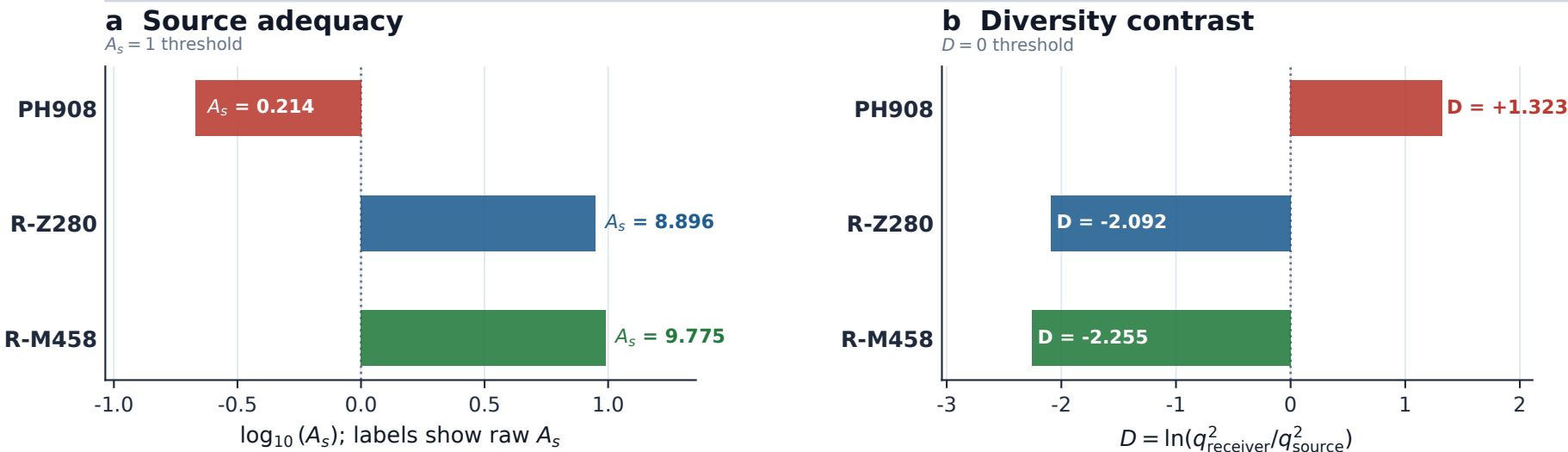


Matched source-receiver checks separate compatible R1a comparators from held-out PH908

Source = Eastern Europe/Russia; receiver = Balkans.

Primary YFull A_s and D bars are shown with branch-unit stability markers and bounded support traces.



d Support and calibration context

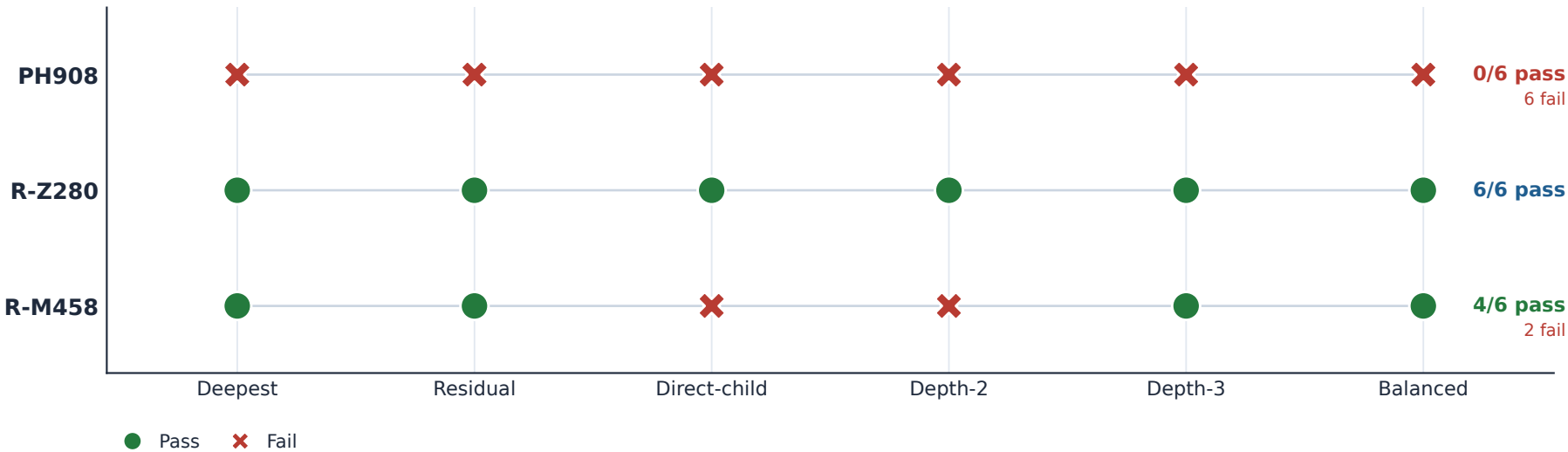
Bounded support layers		
	HRAS/YSEQ A_s / D	FTDNA A_s (D n/a)
● PH908	0.202 / +1.392	0.486; D n/a
● R-Z280	8.242 / -2.113	9.471; D n/a
● R-M458	9.857 / -2.249	17.072; D n/a

Specificity/calibration trace

Separation gap	AUROC
0.303	1.00
Matched-block permutation	T95 negative-control flags
p = 5.0e-5	6/136

FTDNA is bounded direct-child source-adequacy support; calibration metrics are conditional diagnostics, not population-level false-positive rates.

c Branch-unit stability across evaluated resolution layers



Markers show observed pass/fail branch-unit states; no-call is defined in the decision grammar but is not observed in this contrast.

Interpretation boundary: Figure 3 tests a predefined PH908/R1a source-receiver contrast. Branch-unit markers show observed pass/fail stability across audited resolution layers; they are not rarefaction or confidence intervals.

The figure does not identify PH908 geographic origin, ethnic identity, population continuity, migration route, ancient-DNA confirmation, or a population-level false-positive rate.

Sources: Source data are provided with the numbered Figure 3 source-data bundle. Primary bars use YFull rows for I-PH908, R-Z280, and R-M458; branch-unit dots derive from Tables 11 and 12; HRAS/YSEQ and FTDNA Discover values are bounded support layers; calibration values derive from Table 5.